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URL of this page: https://medlineplus.gov/lab-tests/tsh-thyroid-stimulating-hormone-test/

TSH (Thyroid-stimulating hormone) Test

What is a TSH Test?

TSH stands for thyroid stimulating hormone [https://medlineplus.gov/hormones.html] . Another name for it is thyrotropin. A TSH test is a blood test that measures this hormone. A TSH level that is too high or too low may be a sign of a thyroid problem [https://medlineplus.gov/thyroiddiseases.html] .

The thyroid is a small, butterfly-shaped gland in the front of your neck. Your thyroid makes hormones that control how your body uses energy. Thyroid hormones affect nearly every organ and control many of your body's most important functions. For example, they affect your breathing, heart rate, weight, digestion, and even your mood. If you don't have enough thyroid hormones in your blood, many of your body functions will slow down. But if you have too much, many body functions will speed up.

Your thyroid is controlled by a gland in your brain called the pituitary gland. The pituitary gland makes thyroid-stimulating hormone (TSH). TSH tells your thyroid how much thyroid hormone it needs to make.

If the thyroid hormone level in your blood is too low, your pituitary gland makes larger amounts of TSH to tell your thyroid to work harder. If your thyroid hormone level is too high, the pituitary gland makes little or no TSH. By measuring the TSH level in your blood, you can find out if your thyroid is making the right level of hormones.

Other names: thyrotropin test, thyrotropic hormone

What is it used for?

A TSH test is used to find out how well your thyroid is working. It can tell if you have hyperthyroidism [https://medlineplus.gov/hyperthyroidism.html] (too much thyroid hormone) or hypothyroidism [https://medlineplus.gov/hypothyroidism.html] (too little thyroid hormone) in your blood. But a TSH test can't show what is causing a thyroid problem.

If you take prescription thyroid hormone medicine because of hypothyroidism or because you had your thyroid removed, you'll have regular TSH tests to check your thyroid hormone levels. This testing is done to check that you are taking the right dose of the medicine. TSH tests are also used to monitor your thyroid hormone levels after treatment for hyperthyroidism.

Why do I need a TSH test?

You may need a TSH test if you have symptoms of too much or too little thyroid hormone in your blood.

Hyperthyroidism is also called overactive thyroid. Having more thyroid hormones than you need speeds up your body functions and causes symptoms that can vary from person to person. The symptoms may include::

- Weight loss, even though you may be eating more than usual
- Rapid or irregular heartbeat [https://medlineplus.gov/arrhythmia.html]
- Feeling nervous or irritable
- Trouble sleeping [https://medlineplus.gov/insomnia.html]
- Fatigue [https://medlineplus.gov/fatigue.html]
- Shaky hands [https://medlineplus.gov/tremor.html] , muscle weakness
- Sweating [https://medlineplus.gov/sweat.html] or being very sensitive to heat
- Frequent bowel movements (pooping a lot) or diarrhea [https://medlineplus.gov/diarrhea.html]
- Goiter (enlarged thyroid that may cause your neck to look swollen)

Adults over age 60 may have different symptoms than younger adults. For example, they may lose their appetite or withdraw from other people. Sometimes, this can be mistaken for depression [https://medlineplus.gov/depression.html] or dementia [https://medlineplus.gov/dementia.html] .

Hypothyroidism (too little thyroid hormone) is also called underactive thyroid. Not having enough thyroid hormone slows down your body functions and causes symptoms that can vary from person to person. The symptoms may include:

- Fatigue
- Weight gain
- Being very sensitive to cold
- Joint and muscle pain
- Dry skin
- Dry, thinning hair [https://medlineplus.gov/hairloss.html]
- Heavy or irregular menstrual periods [https://medlineplus.gov/menstruation.html]
- Fertility problems in women [https://medlineplus.gov/femaleinfertility.html]
- Depression
- Constipation [https://medlineplus.gov/constipation.html]

Because hypothyroidism develops slowly, many people don't notice symptoms of the disease for months or even years.

You may also have a TSH test and other tests to help diagnose unusual bumps or lumps on your thyroid. The tests will help your health care provider check for thyroid cancer [https://medlineplus.gov/thyroidcancer.html] or thyroid nodules (growths on your thyroid that aren't cancer).

What happens during a TSH test?

A health care professional will take a blood sample from a vein in your arm, using a small needle. After the needle is inserted, a small amount of blood will be collected into a test tube or vial. You may feel a little sting when the needle goes in or out. This usually takes less than five minutes.

Will I need to do anything to prepare for the test?

You may need to stop taking certain medicines before this test, so tell your provider about everything you take. But don't stop taking any medicines unless your provider tells you to. If your health care provider has ordered other blood tests, you may need to fast [https://medlineplus.gov/lab-tests/fasting-for-a-blood-test/] (not eat or drink) for several hours before the test. Your provider will let you know if there are any special instructions.

Are there any risks to the test?

There is very little risk to having a blood test. You may have slight pain or bruising at the spot where the needle was put in, but most symptoms go away quickly.

What do the results mean?

Your TSH test results can tell you if your thyroid is making too much or too little hormones. But the test can't explain why your TSH levels may be too high or too low.

If your test results aren't normal, your provider will probably order other thyroid blood tests [https://medlineplus.gov/thyroidtests.html] to find out what's causing your thyroid problem. These blood tests may include:

- T4 thyroid hormone test [https://medlineplus.gov/lab-tests/thyroxine-t4-test/]
- T3 thyroid hormone test [https://medlineplus.gov/lab-tests/triiodothyronine-t3-tests/]
- Thyroid antibodies test [https://medlineplus.gov/lab-tests/thyroid-antibodies/] to help diagnose an autoimmune [https://medlineplus.gov/autoimmunediseases.html] thyroid disorder, such as:
    - Graves' disease, the most common cause of hyperthyroidism
    - Hashimoto's disease, that the most common cause of hypothyroidism

In certain cases, an abnormal TSH result may be a sign of a pituitary gland problem [https://medlineplus.gov/pituitarydisorders.html] , but this doesn't happen often.

Learn more about laboratory tests, reference ranges, and understanding results [https://medlineplus.gov/lab-tests/how-to-understand-your-lab-results/] .

Is there anything else I need to know about a TSH test?

Your TSH level may be high or low even when your thyroid gland is healthy. For example, the following can affect your TSH level:

- Some serious illnesses (that are not related to your thyroid) may cause a lower TSH level for a short time.

- The TSH level may be higher in people over age 80, even though they don't have any thyroid problems.
- Pregnancy also affects your TSH level. They are often a little low during the first three months. But sometimes, thyroid disease develops during pregnancy. If you develop thyroid disease during pregnancy, your provider will monitor your condition throughout your pregnancy and after your baby is born. That's because hyperthyroidism, and less often, hypothyroidism, can continue after you give birth. If you have a history of thyroid disease, be sure to talk with your provider if you are pregnant or are thinking of becoming pregnant.

Newborn babies are routinely screened with a TSH test to see if they were born with hypothyroidism (congenital hypothyroidism).

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